

Smart Metro Ticket Booking System

Requirement Analysis Phase

Team ID

Project Name: Smart Metro Booking System

Maximum Marks: 4 Marks

4. Technology Stack

This document sets out the technology stack proposed for the ServiceNow-based Metro Ticket Booking system, covering the presentation, application logic, integration, data, notification, and security layers.

Figure 4.1 — Layered technology architecture of the ticketing solution

4.1 Layer-wise Technology Breakdown

Layer	Technology / Component	Purpose
Presentation	ServiceNow Service Portal, UI Pages, WhatsApp Business Chatbot	Passenger-facing booking form, fare preview, ticket display, and chat-based booking.
Application / Process	Service Catalog, Flow Designer, Business Rules, Script Includes, Catalog Client Scripts	Captures booking input, orchestrates the booking-to-ticket workflow, executes fare logic.
Integration	Integration Hub, REST/SOAP APIs, Payment Gateway (UPI/Card/Wallet), QR Code Generation Library	Connects the platform to external payment and QR services with logged, traceable calls.
Data	ServiceNow Custom Tables — Station Master, Fare Rule Table, Ticket/QR Records, Transaction Logs	Stores reference and transactional data needed for booking, validation, and reporting.
Notification	Email Notifications, SMS Gateway, Push Notifications, WhatsApp Business API	Delivers QR tickets, payment confirmations, and exception alerts to passengers and staff.
Reporting & Analytics	Performance Analytics, Reports, Dashboards	Surfaces booking volume, revenue split, and peak-time insights for Metro Operations.
Security & Administration	Access Control Lists (ACLs), OAuth 2.0, Data Encryption, Role-based Views	Restricts data and functionality by persona and secures payment/QR data end-to-end.

4.2 Platform Rationale

- Service Catalog offers a governed, reusable form framework for capturing structured booking input (Source, Destination, Passenger Type, Number of Tickets) with built-in client-side validation.
- Flow Designer enables low-code, visual orchestration of the multi-step booking-to-ticket process, keeping the workflow easy to audit and change without heavy scripting.
- Integration Hub standardizes outbound calls to the payment gateway and QR service, giving IT Admins centralized logging, retry handling, and credential management.
- Native Notifications keep ticket delivery inside the ServiceNow platform, avoiding a separate messaging system while still supporting Email, SMS, Push, and WhatsApp.
- Performance Analytics and Reports reuse existing platform capability to deliver the data-driven insights Metro Operations needs for route and peak-time planning, without a separate BI tool.

4.3 Environment Considerations

The solution runs entirely on the ServiceNow platform (Now Platform), using out-of-the-box Service Portal capability for the front end and Integration Hub spokes for third-party connections. This keeps the technology footprint small, avoids licensing overhead for a separate mobile app in the first phase, and lets every persona — Passengers, Station Managers, Metro Operations, and IT Admins — be served from one centrally administered

system.

Figure 4.2 — Single-platform deployment model serving all four personas